Story Notes: Climate and Influenza A HA Evolution

Key findings, methods sketch, and open questions

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Pipeline overview

The analysis takes GISAID Influenza A sequences and matched ERA5-Land climate data and asks whether environmental conditions predict molecular evolution of the haemagglutinin (HA) surface protein.

Sequences and metadata. Sequences for H1N1 (H1, N1) and H3N2 (H3, N2) surface proteins plus PA (internal protein, negative control) were exported from GISAID EpiFlu. Metadata were geocoded via Nominatim and monthly ERA5-Land temperature and relative humidity (derived from dewpoint via Magnus formula) were attached to each sequence by collection location and date.

Phylogenetics. Sequences were aligned with MAFFT, codon-aligned with MACSE (pairwise batch approach), and initial trees built with MAPLE. Two-round pruning removed duplicate/identical sequences (zero-length branches) and outliers (unusually long branches). Trees were dated with TreeTime to obtain a timetree with branch lengths in years.

Evolutionary rates. Two parallel approaches were used:

1. *MAPLE/timetree branch rate:* total substitutions per year on each terminal branch, as estimated directly from the timetree (`log_rate`).
2. *Ancestral parsimony reconstruction:* for each codon in the alignment, ancestral states were reconstructed at internal nodes via maximum parsimony (`phangorn::ancestral.pars`, MPR). Non-synonymous (NS) and synonymous (S) changes on each terminal branch were counted and divided by timetree branch length to give per-year rates. This was applied to three regions: the 7-codon antigenic ridge, the full epitope (∼103 codons), and the whole HA gene (∼579 codons). The approach avoids the dS denominator problem that makes yn00-based ω unreliable on short or sparse alignments.
3. *yn00 ω (dN/dS):* pairwise yn00 estimates mapped onto the tree via NNLS and FastME. Used for epitope-level analyses but largely superseded by ancestral parsimony; results were null after applying appropriate rate floors.

Climate predictors. All predictors were scaled (mean 0, SD 1): temperature (`scaled_temp`), relative humidity (`scaled_humidity`), Δtemperature and Δhumidity (change from previous sampling interval), sampling density (to absorb seasonal confounding), and population density (not yet fully integrated; largely NA).

Statistical models. Phylogenetic regression via `phylolm` (Brownian motion covariance) was used to test climate predictors against log-transformed rates, controlling for year as a factor and for phylogenetic non-independence. Sampling density was included as a covariate to absorb the seasonal flu/sampling winter peak, so surviving temperature signals reflect environmental

effects beyond seasonality. BH-FDR correction was applied across all predictor × response combinations within each gene. A Wilcoxon rank-sum test with rank-biserial effect size was used to test whether specific recurrent mutations (>5 independent occurrences) arise preferentially under particular climate conditions.

Key findings

1. Faster HA evolution in colder climates

This is the central finding and it is **consistent across methods, genes, and genomic regions**.

- **MAPLE/timetree branch rate (total substitutions/year):** Temperature has a significant negative effect on `log_rate` in both H1N1 (estimate = -0.047, BH-FDR $p = 3 \times 10^{-4}$) and H3N2 (estimate = -0.030, BH-FDR $p = 0.016$) after controlling for sampling density, year, and humidity.
- **Ancestral parsimony — full gene NS rate:** H1N1 estimate = -0.059 (FDR $p = 4 \times 10^{-6}$); H3N2 estimate = -0.041 (FDR $p = 4.9 \times 10^{-3}$).
- **Ancestral parsimony — full gene S rate:** H1N1 estimate = -0.099 (FDR $p = 4 \times 10^{-6}$); H3N2 estimate = -0.035 (FDR $p = 6.4 \times 10^{-3}$). The effect on synonymous rates — which are largely neutral — is noteworthy: it points to a *mutational* rather than purely selective mechanism.
- **Epitope NS rate:** H1N1 estimate = -0.106 (FDR $p = 6.5 \times 10^{-5}$); H3N2 (marginal, FDR $p \approx 0.10$).
- **Ridge NS rate:** H1N1 estimate = -0.186 (FDR $p = 0.014$). Weaker signal, consistent direction.

Interpretation. The negative temperature coefficient means lower temperatures are associated with higher per-branch evolutionary rates. This aligns with the winter seasonality of influenza: cold conditions coincide with peak transmission, faster replication, and potentially elevated polymerase error rates. The fact that both NS and S rates are affected, and that the effect survives after controlling for sampling density (which absorbs the seasonal flu peak), supports a genuine environmental effect on the HA mutation rate rather than a sampling artefact.

Sampling density as a control. Sampling density is positively associated with rates in several analyses (particularly H3N2, FDR $p < 10^{-4}$), consistent with denser sampling capturing more short-branch diversity. Its inclusion is necessary; temperature results should not be interpreted from models that omit it.

2. No specific recurrent mutation is climate-associated

Wilcoxon rank-sum tests were applied to all recurrent NS mutations ($n > 5$ independent terminal-branch occurrences) in the antigenic ridge and full epitope, testing association with temperature, humidity, Δtemperature, and Δhumidity.

After BH-FDR correction, **no mutation in the ridge survived** in either H1N1 or H3N2. In the epitope, a single H3N2 mutation at residue 47 (H47R, $n = 9$, rank-biserial $r = -0.73$, FDR $p = 0.044$) was significant for temperature; this is a single finding among hundreds of tests and should be treated cautiously pending replication.

Interpretation. The climate effect on HA evolution appears to operate on the *overall rate* rather than driving specific adaptive substitutions at identifiable positions. This is consistent

with a mutational (polymerase fidelity) rather than a selective (immune-driven fixation) mechanism.

3. INLA results not reliable

INLA (Integrated Nested Laplace Approximation) was tested as an alternative Bayesian hierarchical approach. Results were found to be **sensitive to the random seed used for subset sampling**: different seeds produced qualitatively different conclusions, indicating that the INLA samples were not representative of the full posterior. This likely reflects the large dataset size and the complexity of the spatial and phylogenetic random effects. INLA results are excluded from the final analysis; `phylolm` is used throughout.

4. Additional notes

- **Ridge ω artefact.** Early analyses showed an apparent humidity $\rightarrow$ ridge ω association. This was traced to a dS denominator problem: with only 7 codons, $\approx 96\%$ of ridge dS values are effectively zero, making $\omega = \text{dN/dS}$ undefined or inflated. The result disappears when ancestral parsimony rates are used instead. A reminder that ω is not interpretable on very short alignments.
- **Null result for yn00 ω after rate floors.** After applying log-rate floors to remove implausibly small dN and dS estimates (bimodal gap in distributions), all epitope-level ω phylolm associations became non-significant (except the ridge artefact above). The ancestral parsimony approach is therefore preferred for all substitution-rate analyses.
- **PA negative control.** PA (polymerase acidic, internal protein) serves as a negative control. PA should show no climate signal if the HA findings reflect surface-protein-specific biology. Results for PA were not assessed in the epitope/ridge pipeline (surface-protein-specific) but should be checked in the main phylolm results before finalising.
- **Geographic confounding.** Sequences are dominated by temperate Northern Hemisphere sampling (US, Europe). Temperature is partly a proxy for geography/latitude. Phylolm corrects for phylogenetic clustering but geographic and phylogenetic structure are correlated in flu. This should be acknowledged as a limitation.
- **Population density.** The `scaled_pop_density` predictor is largely NA in the current dataset (source not yet integrated). A significant negative effect on H1N1 ridge NS rate was observed in the data that do have values; interpret cautiously.

Open questions / next steps

- Check PA negative control in main phylolm results.
- Verify H47R (epitope, H3N2) in independent dataset or with larger n.
- Consider latitude or hemisphere as an explicit covariate to disentangle temperature from geography.
- Integrate population density data to complete the predictor set.
- Investigate whether the S-rate temperature effect differs between coding regions (would help distinguish polymerase fidelity vs. replication rate hypotheses).